

JEE Mains 2019 Chapter wise Question Bank

Trigonometrical Ratios – Questions

Q1

For any $\theta \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ the expression

$3(\sin\theta - \cos\theta)^4 + 6(\sin\theta + \cos\theta)^2 + 4\sin^6\theta$ equals:

- (1) $13 - 4\cos^2\theta + 6\sin^2\theta\cos^2\theta$
 (2) $13 - 4\cos^6\theta$
 (3) $13 - 4\cos^2\theta + 6\cos^4\theta$
 (4) $13 - 4\cos^4\theta + 2\sin^2\theta\cos^2\theta$

9 Jan Morning

Q2

The value of

$$\cos \frac{\pi}{2^2} \cdot \cos \frac{\pi}{2^3} \cdot \dots \cdot \cos \frac{\pi}{2^{10}} \cdot \sin \frac{\pi}{2^{10}}$$

- (1) $\frac{1}{512}$ (2) $\frac{1}{1024}$
 (3) $\frac{1}{256}$ (4) $\frac{1}{2}$

10 Jan Evening

Q3

Let $f_k(x) = \frac{1}{k}(\sin^k x + \cos^k x)$ for $k = 1, 2, 3, \dots$. Then for

all $x \in \mathbb{R}$, the value of $f_4(x) - f_6(x)$ is equal to :

- (1) $\frac{1}{12}$ (2) $\frac{1}{4}$
 (3) $-\frac{1}{12}$ (4) $\frac{5}{12}$

11 Jan Morning

Q4

If $\sin^4\alpha + 4\cos^4\beta + 2 = 4\sqrt{2} \sin\alpha \cos\beta$; $\alpha, \beta \in [0, \pi]$, then $\cos(\alpha + \beta) - \cos(\alpha - \beta)$ is equal to :

- (1) 0 (2) -1
 (3) $\sqrt{2}$ (4) $-\sqrt{2}$

12 Jan Evening

Q5

If $\cos(\alpha + \beta) = \frac{3}{5}$, $\sin(\alpha - \beta) = \frac{5}{13}$ and $0 < \alpha, \beta < \frac{\pi}{4}$, then

$\tan(2\alpha)$ is equal to :

- (1) $\frac{63}{52}$ (2) $\frac{63}{16}$ (3) $\frac{21}{16}$ (4) $\frac{33}{52}$

8 April Morning

Q6

The value of

$$\cos^2 10^\circ - \cos 10^\circ \cos 50^\circ + \cos^2 50^\circ$$
 is :

- (1) $\frac{3}{4} + \cos 20^\circ$ (2) $3/4$
 (3) $\frac{3}{2}(1 + \cos 20^\circ)$ (4) $3/2$

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Q7

Let $S = \{\theta \in [-2\pi, 2\pi] : 2\cos^2\theta + 3\sin\theta = 0\}$.

Then the sum of the elements of S is:

- (1) $\frac{13\pi}{6}$ (2) $\frac{5\pi}{3}$ (3) 2π (4) π

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Q8

The value of $\sin 10^\circ \sin 30^\circ \sin 50^\circ \sin 70^\circ$ is:

- (1) $\frac{1}{16}$ (2) $\frac{1}{32}$ (3) $\frac{1}{18}$ (4) $\frac{1}{36}$

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Trigonometrical Ratios - Solutions

Q1

$$\begin{aligned}
 \cdot (2) \quad & 3(\sin \theta - \cos \theta)^4 + 6(\sin \theta + \cos \theta)^2 + 4\sin^6 \theta \\
 &= 3(1 - 2\sin \theta \cos \theta)^2 + 6(1 + 2\sin \theta \cos \theta) + 4\sin^6 \theta \\
 &= 3(1 + 4\sin^2 \theta \cos^2 \theta - 4\sin \theta \cos \theta) + 6 \\
 &\quad + 12\sin \theta \cos \theta + 4\sin^6 \theta \\
 &= 9 + 12\sin^2 \theta \cos^2 \theta + 4\sin^6 \theta \\
 &= 9 + 12\cos^2 \theta(1 - \cos^2 \theta) + 4(1 - \cos^2 \theta)^3 \\
 &= 9 + 12\cos^2 \theta - 12\cos^4 \theta \\
 &\quad + 4(1 - \cos^6 \theta - 3\cos^2 \theta + 3\cos^4 \theta) \\
 &= 9 + 4 - 4\cos^6 \theta \\
 &= 13 - 4\cos^6 \theta
 \end{aligned}$$

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Q2

$$\begin{aligned}
 (1) \quad A &= \cos \frac{\pi}{2^2} \cdot \cos \frac{\pi}{2^3} \dots \cos \frac{\pi}{2^{10}} \cdot \sin \frac{\pi}{2^{10}} \\
 &= \frac{1}{2} \left(\cos \frac{\pi}{2^2} \cdot \cos \frac{\pi}{2^3} \dots \cos \frac{\pi}{2^9} \sin \frac{\pi}{2^9} \right) \\
 &= \frac{1}{2^8} \left(\cos \frac{\pi}{2^2} \cdot \sin \frac{\pi}{2^2} \right) = \frac{1}{2^9} \sin \frac{\pi}{2} \\
 &= \frac{1}{512}
 \end{aligned}$$

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Q3

$$(1) \quad f_k(x) = \frac{1}{k}(\sin^k x + \cos^k x)$$

$$\begin{aligned}
 f_4(x) &= \frac{1}{4}[\sin^4 x + \cos^4 x] \\
 &= \frac{1}{4} \left[(\sin^2 x + \cos^2 x)^2 - \frac{(\sin 2x)^2}{2} \right] \\
 &= \frac{1}{4} \left[1 - \frac{(\sin 2x)^2}{2} \right]
 \end{aligned}$$

$$\begin{aligned}
 f_6(x) &= \frac{1}{6}[\sin^6 x + \cos^6 x] \\
 &= \frac{1}{6} \left[(\sin^2 x + \cos^2 x) - \frac{3}{4}(\sin^2 x)^2 \right]
 \end{aligned}$$

$$= \frac{1}{6} \left[1 - \frac{3}{4}(\sin 2x)^2 \right]$$

$$\begin{aligned}
 \text{Now } f_4(x) - f_6(x) &= \frac{1}{4} - \frac{1}{6} - \frac{(\sin 2x)^2}{8} + \frac{1}{8}(\sin 2x)^2 \\
 &= \frac{1}{12}
 \end{aligned}$$

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Q4

Trigonometrical Ratios

(4) $\therefore$ The given equation is

$$\sin^4 \alpha + 4 \cos^4 \beta + 2 = 4\sqrt{2} \sin \alpha \cdot \cos \beta,$$

$$\alpha, \beta \in [0, \pi]$$

Then, by A.M., G.M. inequality;

$$\text{A.M.} \geq \text{G.M.}$$

$$\frac{\sin^4 \alpha + 4 \cos^4 \beta + 1 + 1}{4} \geq (\sin^4 \alpha \cdot 4 \cos^4 \beta \cdot 1 \cdot 1)^{\frac{1}{4}}$$

$$\sin^4 \alpha + 4 \cos^4 \beta + 1 + 1 \geq 4\sqrt{2} \sin \alpha \cdot |\cos \beta|$$

Inequality still holds when $\cos \beta < 0$ but L.H.S. is positive than $\cos \beta > 0$, then

$$\text{L.H.S.} = \text{R.H.S.}$$

$$\therefore \sin^4 \alpha = 1 \text{ and } \cos^4 \beta = \frac{1}{4}$$

$$\Rightarrow \alpha = \frac{\pi}{2} \text{ and } \beta = \frac{\pi}{4}$$

$$\therefore \cos(\alpha + \beta) - \cos(\alpha - \beta)$$

$$= \cos\left(\frac{\pi}{2} + \beta\right) - \cos\left(\frac{\pi}{2} - \beta\right)$$

$$= -\sin \beta - \sin \beta = -2 \sin \frac{\pi}{4} = -\sqrt{2}$$

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Q5

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(2) $\therefore \alpha + \beta$ and $\alpha - \beta$ both are acute angles.

$$\cos(\alpha + \beta) = \frac{3}{5}, \text{ then } \sin(\alpha + \beta) = \sqrt{1 - \left(\frac{3}{5}\right)^2} = \frac{4}{5}$$

$$\Rightarrow \tan(\alpha + \beta) = \frac{4}{3}$$

$$\text{And } \sin(\alpha - \beta) = \frac{5}{13}, \text{ then}$$

$$\cos(\alpha - \beta) = \sqrt{1 - \left(\frac{5}{13}\right)^2} = \frac{12}{13}$$

$$\Rightarrow \tan(\alpha - \beta) = \frac{5}{12}$$

$$\text{Now, } \tan 2\alpha = \tan((\alpha + \beta) + (\alpha - \beta))$$

$$= \frac{\tan(\alpha + \beta) + \tan(\alpha - \beta)}{1 - \tan(\alpha + \beta) \cdot \tan(\alpha - \beta)}$$

$$= \frac{\frac{4}{3} + \frac{5}{12}}{1 - \frac{4}{3} \cdot \frac{5}{12}} = \frac{63}{16}$$

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Q6

$$(2) \cos^2 10^\circ - \cos 10^\circ \cos 50^\circ + \cos^2 50^\circ$$

$$= \left(\frac{1 + \cos 20^\circ}{2}\right) + \left(\frac{1 + \cos 100^\circ}{2}\right) - \frac{1}{2}(2 \cos 10^\circ \cos 50^\circ)$$

$$= 1 + \frac{1}{2}(\cos 20^\circ + \cos 100^\circ) - \frac{1}{2}[\cos 60^\circ + \cos 40^\circ]$$

$$= \left(1 - \frac{1}{4}\right) + \frac{1}{2}[\cos 20^\circ + \cos 100^\circ - \cos 40^\circ]$$

$$= \frac{3}{4} + \frac{1}{2}[2 \cos 60^\circ \times \cos 40^\circ - \cos 40^\circ]$$

$$= \frac{3}{4}$$

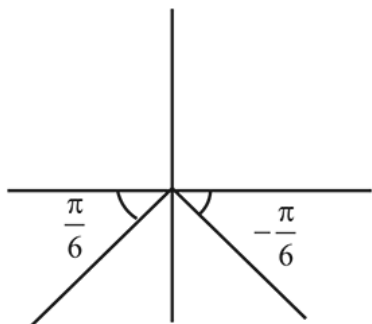
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Q7

$$(3) \quad 2\cos^2\theta + 3\sin\theta = 0$$

$$\Rightarrow (2\sin\theta + 1)(\sin\theta - 2) = 0$$

$$\Rightarrow \sin\theta = -\frac{1}{2} \text{ or } \sin\theta = 2 \rightarrow \text{Not possible}$$



The required sum of all solutions in $[-2\pi, 2\pi]$ is

$$= \left(\pi + \frac{\pi}{6}\right) + \left(2\pi - \frac{\pi}{6}\right) + \left(-\frac{\pi}{6}\right) + \left(-\pi + \frac{\pi}{6}\right) = 2\pi$$

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Q8

$$(1) \quad \because \sin(60^\circ + A) \cdot \sin(60^\circ - A) \sin A = \frac{1}{4} \sin 3A$$

$$\therefore \sin 10^\circ \sin 50^\circ \sin 70^\circ = \sin 10^\circ \sin(60^\circ - 10^\circ)$$

$$\sin(60^\circ + 10^\circ) = \frac{1}{4} \sin 30^\circ$$

$$\Rightarrow \sin 10^\circ \sin 30^\circ \sin 50^\circ \sin 70^\circ = \frac{1}{4} \sin^2 30^\circ = \frac{1}{16}$$

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